Introduction To Image Processing

What is Digital Image Processing?

An image may be defined as a two-dimensional function, f(x,y), where x and y are spatial (plane) coordinates, and the amplitude of f at any pair of coordinates (x, y) is called the intensity or gray level of the image at that point.

When x, y, and the intensity values of f are all finite, discrete quantities, we call the image a digital image.

Pixel (picture element, *image element*, *pel*, and *pixel)* is the term used most widely to denote the elements of a digital image .

What is Digital Image Processing?

How many types of wavelengths are there?

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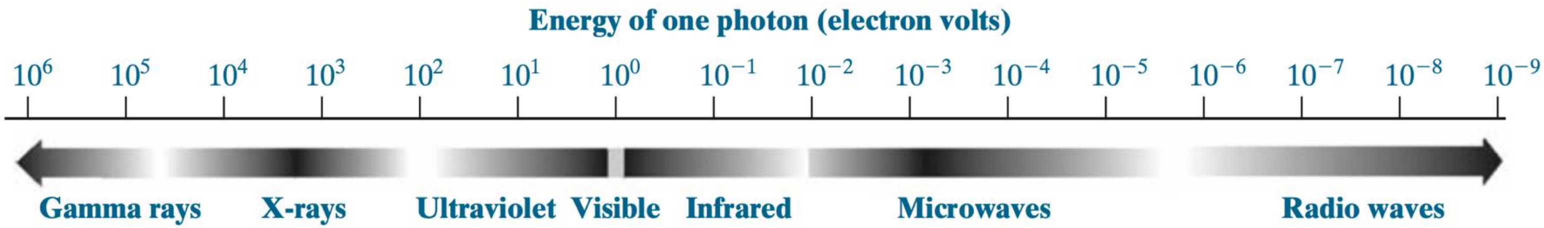

The electromagnetic spectrum arranged according to energy per photon

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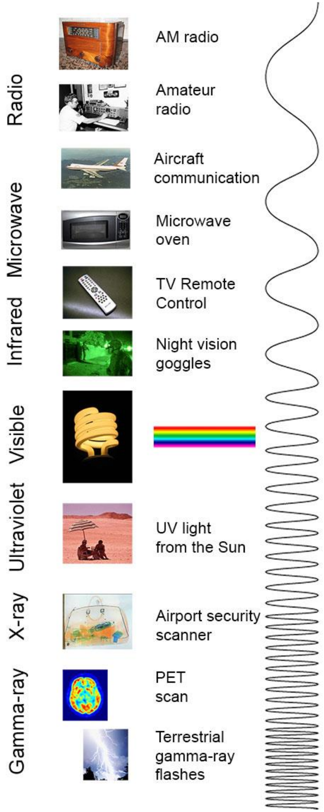

Radio wave: AM Radio, FM Radio, Cellular Networks,…

Microwave: Microwave Ovens, Satellite Communication,…

Infraredwave: Remote Controls, Thermal Imaging, Heating,…

Visible: Colors, Lighting, Photography,…

Ultraviolet: UVA, UVB, UVC, Medical Applications, Tanning,…

X-ray: Medical Imaging, Dental X-rays, CT Scans, Security Screening,...

Gamma-ray: Medical Applications, Astrophysics, Industrial Applications,…

https://imagine.gsfc.nasa.gov/science/toolbox/emspectrum1.html

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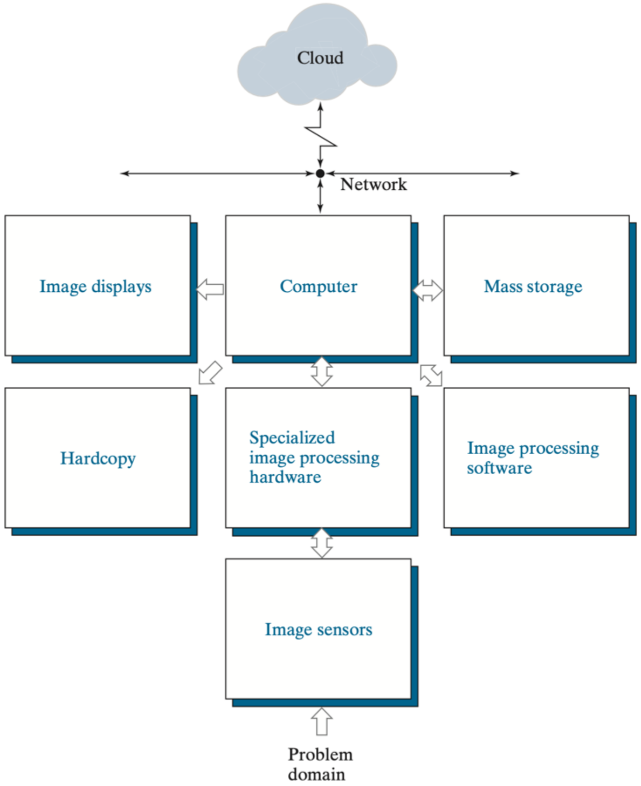

Components of a general-purpose image processing system.

Fundamental Steps in Digital Image Processing

- Image acquisition
- Image enhancement
- Image restoration
- Color image processing
- Wavelets
- Compression
- Morphological
- Segmentation